Splinter: A Lock-Free Shared-Memory Substrate For Tightly-Coupled Inference And Governance

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Abstract

Traditional inter-process communication (IPC) and database architectures frequently suffer from "socket tax" — the cumulative latency of kernel interrupts, context switching, and serialization. We introduce **Splinter**, a minimalist C-based key-value manifold designed as a passive shared-memory substrate. By utilizing a lock-free, 64-byte aligned architecture, Splinter achieves mechanical sympathy (Drepper 2007) with the Linux virtual memory manager and modern CPU cache hierarchies. This enables direct pointer access to what used to be a `memcpy()` operation, permitting observational semantic governance of inference output in near real-time. We further describe a cooperative memory scheduling protocol for loadable Logic Shards, and examine the architecture's implications for compliance with AI accountability and risk management frameworks.

Introduction

In the landscape of local Large Language Model (LLM) runtimes, the primary bottleneck has shifted from raw computation to the "last millimeter" of data coordination. Standard tools like Redis or SQLite, while architecturally robust, function as active orchestrators requiring significant overhead for simple state synchronization. More critically for governance applications, they introduce an irreducible observation gap: output must be serialized and transmitted before it can be inspected, meaning noncompliant content exists — however briefly — in a state that oversight cannot yet see.

Splinter is proposed as a **passive substrate**. It rejects the active-server model in favor of a memory-mapped manifold where processes swim in the same address space. This approach eliminates the need for a central arbiter, allowing for unmediated, zero-copy access to 768-dimensional vectors and metadata. The governor and the actor share the same physical RAM, which means the observation gap closes by construction rather than by policy.

This property is not theoretical. The `evProcessor` reference implementation demonstrates a working semantic governance bus: fifteen emotional and epistemic dimensions with graduated severity levels — ambient, personal-stake,

and communal-scale signals — fully observable by any process on the manifold without data copying. Governance in Splinter is not a layer bolted onto an existing architecture. It is a mechanical consequence of the address space design.

Design and Implementation

The overarching goal in Splinter's design was eschewing any feature that would require computation during a write that was not confined to bitwise operations. All computation in Splinter is offloaded to the included client, which manages its own heap.

Logic Shards, planned to ship mid-2026, allow individual consumers to coordinate POSIX memory advisement with univocal signal on intentions within the shared region using a self-electing, voluntarily-yielding algorithm described in full in Section 4. This prevents individual workers from inadvertently confusing the kernel by causing it to constantly re-evaluate its page cache strategy for the same region of memory.

Once in the client, callers are free to run any transformations or triggers they require. WASM and Lua are included as examples for self-executing keys arranged purely on the client side.

Escaping The "Socket And Lock" (S&L) Tax

Splinter Is A Unified Semantic Backplane.

In the Splinter architecture, we reject the notion that an AI agent is a standalone black box. Instead, we propose an ecosystem where Inference, Observation, and Governance inhabit a single, 64-byte aligned address space.

By utilizing lock-free atomics and signal-group resonance, we eliminate the Socket Tax that traditionally separates thought from oversight. This is not merely a performance optimization; it is a fundamental shift in how semantic control is expressed architecturally.

When the Governor and the Actor swim in the same L3 cache pool, the feedback loop for safety becomes instantaneous. We move from post-hoc validation to **Prospective Integrity** — detecting the divergence of an agent's internal belief state before the first bit of a high-risk token sequence can be serialized to any downstream consumer.

Mechanical Sympathy and Alignment

The core of Splinter's performance is its adherence to the physical constraints of the CPU. Every data slot is aligned to a 64-byte boundary, matching the standard L1 cache line size.

- **Atomic Harmony:** By isolating keys into distinct cache lines, Splinter prevents false sharing, ensuring that multiple cores do not invalidate each

other's cache while writing to adjacent memory.

- **The Hybrid Mop:** To optimize I/O, Splinter employs a hygiene strategy that zeroes out only the 64-byte tail of a write. This signals the kernel's page cache that a full line is dirty, encouraging efficient burst writes to disk without the cost of zeroing the entire slot allocation.

Disjointed-Lane Collision Protocol

Splinter utilizes a (Fraser 2004)lock-free linear probing strategy to resolve hash collisions:

$(H(k) \bmod N)$. The implementation ensures atomicity by utilizing `compare_and_swap` (CAS) operations on the slot hash identifier:

$$i = H(k) \bmod N$$

If a collision is detected, the writer probes the next 64-byte aligned lane until an empty or matching slot is identified. This allows for a Multi-Reader/Multi-Writer (MRMW) environment without global locks.

The seqlock protocol enforces consistency across this environment. Every slot carries a 64-bit epoch counter. An even epoch indicates a stable slot, safe to read or write. An odd epoch indicates a writer is active. Readers validate the epoch before and after every read; a changed epoch means the snapshot is torn and must be retried. This protocol requires no kernel involvement and no mutex acquisition on any path.

Performance Evaluation

Splinter is not a general-purpose data store. It is a coordination substrate — a fast lane that sits in front of systems like Redis, SQLite, or any socket-bound store, handling the state that needs to be visible across processes *before* it is worth persisting or routing elsewhere. Comparing Splinter's throughput against Redis is therefore a category error: it is like benchmarking a CPU register file against a disk cache. They are coupled components in a pipeline, not competitors for the same role.

What Splinter eliminates is the *entry cost* of coordination: the kernel interrupt, the context switch, the serialize-transmit-deserialize cycle that every socket-based system pays regardless of how fast the store itself is. On inference workloads, this cost is paid on every token boundary, every embedding update, every governance check. At generation speeds, that overhead is not negligible — it is the bottleneck.

Cycles Per Operation (CPO)

We characterize Splinter's efficiency through Cycles Per Operation (CPO):

$$CPO = \frac{f}{Ops}$$

where f is the CPU clock frequency and Ops is sustained throughput. On a Tiger Lake i3 baseline — a modest, thermally-constrained mobile part — Splinter sustains over 10M operations per second, yielding a CPO of approximately 300. This is not a cherry-picked result; it is the floor, measured on hardware that most local inference deployments will exceed.

On modern high-bandwidth architectures (Zen 4/5), where the manifold fits entirely within L3 cache, projected CPO drops to approximately 10 — a figure that puts coordination overhead below the noise floor of any inference workload we are aware of.

NUMA Affinity and Sympathetic Pairing

CPO figures should be understood as baseline single-node measurements. In multi-socket or chiplet configurations, proper NUMA affinity — binding the manifold and all participating processes to the same memory domain via `mbind()` — is the single highest-leverage configuration change available. The `splinter_open_numa()` path exists precisely for this case.

The relationship is multiplicative, not additive: a manifold pinned to the wrong NUMA node will not merely be slower, it will produce inconsistent latency that defeats the purpose of lock-free design. Affinity should be established first and verified before interpreting any throughput measurement.

What The Numbers Do Not Capture

The CPO metric intentionally excludes the cost Splinter does *not* pay: there is no socket round-trip to amortize, no mutex acquisition in the hot path, no serialization format to decode. In a coupled deployment — Splinter handling live inference state, Redis handling durable session storage downstream — the meaningful benchmark is end-to-end governance latency from token emission to policy decision. That number is architecture-dependent and workload-dependent, but the Splinter contribution to it is bounded by the CPO figures above.

Logic Shards and Cooperative Memory Governance

As the number of processes participating in a Splinter deployment grows, a new coordination problem emerges that is distinct from the per-slot seqlock discipline. Individual processes — inference engines, embedding sidecars, governance observers, maintenance workers — have fundamentally different memory access patterns. An embedding backfill sweep wants `POSIX_MADV_SEQUENTIAL`. A live inference loop wants `POSIX_MADV_WILLNEED` on a small hot set of slots. A purge pass may want `POSIX_MADV_DONTNEED` on cold regions. Issued independently,

these advisements produce contradictory signals to the kernel's page cache manager, forcing constant re-evaluation of the same memory region and degrading the cache residency that the entire architecture depends on.

The solution is not to take CPU scheduling away from the OS. Linux and cgroups are already well-equipped for process priority management, and reinventing the scheduler without a precise definition of success over the existing OS scheduler would add complexity without a measurable payoff. The problem is specifically one of *memory intent coordination*, and the solution is scoped accordingly.

Shards

A Logic Shard is a loadable module — a standard shared object conforming to a symbol map defined in `splinter.h` — that can be attached to a running Splinter deployment without modifying the library or its tooling. Shards may implement sidecar inference, semantic analysis, backfill operations, protocol bridges, or any other workload that needs direct manifold access with its own linkage requirements. They are the natural unit of memory intent declaration.

The shard architecture is designed around a hard ceiling of 32 simultaneously loaded shards. This is not an arbitrary limit — it falls directly out of the epoch spinlock geometry. The seqlock protocol guarantees forward progress for at most 32 concurrent writers before the odd-epoch contention window becomes problematic. The shard ceiling and the writer ceiling are the same ceiling, which means the cooperative scheduling table never needs to be larger than the concurrency the protocol already supports.

The Bid Table

Cooperative memory scheduling state lives in a fixed region of the Splinter header. Each of the 32 shard slots contains a bid record:

```
struct splinter_shard_bid {
    // 0 = empty slot
    atomic_uint_least32_t shard_id;
    // WILLNEED / SEQUENTIAL / RANDOM (DONTNEED Is special-cased)
    atomic_uint_least8_t  intent;
    // 0-255, higher wins election
    atomic_uint_least8_t  priority;
    // declared window in TSC ticks
    atomic_uint_least64_t duration_tsc;
    // splinter_now() at registration
    atomic_uint_least64_t claimed_at;
};
```

32 bid records occupy approximately 1KB of header space — negligible relative to the manifold itself. A shard claims a slot on load via CAS on `shard_id`,

declares its intent and duration upfront, and releases the slot on unload or voluntary yield. There is no dynamic allocation and no central arbiter.

The Election

Sovereignty — the right to issue the current `posix_madvise()` call — is determined by a read-only scan of all 32 bid slots. The rule is simple: the highest-priority unexpired bid wins. Ties are broken by `claimed_at`, with earlier registration taking precedence. A bid is considered expired when:

$$splinter_now() - claimed_at > duration_tsc$$

This scan is O(32), which is effectively O(1). It requires no lock and no kernel involvement. Any shard can determine the current sovereign at any time by performing the same scan independently — the result is deterministic from the static bid data alone.

When a shard's declared window expires, it re-bids if it needs more time. This upfront commitment with explicit re-bid is a deliberate fairness guarantee: a shard cannot hold sovereignty indefinitely. A shard that consistently over-commits its declared duration without consuming it creates a measurable record in the TSC timestamps — `claimed_at` and actual yield time are both observable. A feedback loop that penalizes chronic over-committers is a natural future extension; the data required to build it already exists in the bid table.

The `splinter_now()` function already exists in the library for write-latency backfill in the inference sidecars. Its reuse here as the scheduling heartbeat is intentional — the cooperative scheduling primitive costs nothing new in terms of instruction path complexity or syscall overhead.

Voluntary Yield and `splinter_madvise()`

A shard that wants to advise the kernel calls `splinter_madvise()` rather than `posix_madvise()` directly. The call performs the election scan, determines whether the caller is currently sovereign, and either issues the advisement or blocks until the current sovereign's window expires and the caller wins a subsequent election.

The blocking path is a TSC-polled sleep — no futex, no kernel wait. A shard that cannot tolerate blocking can check sovereignty first and defer its work rather than waiting. This is the appropriate pattern for latency-sensitive paths like live inference, which should declare `WILLNEED` with a short window and re-bid frequently rather than holding sovereignty across a long generation pass.

Sharp Edges

`POSIX_MADV_DONTNEED` is a hostile advisement in an active deployment. A shard issuing DONTNEED on a region that active readers depend on will cause page

faults and latency spikes that no priority scheme fully mitigates. DONTNEED bids should be restricted to maintenance shards operating during known quiet periods. A soft bumper — a wrapper that checks for active WILLNEED or SEQUENTIAL bids before permitting a DONTNEED election win — is a natural first contribution for anyone looking to harden the protocol.

The bid table has no authentication. A shard with bus access can claim any priority level. As with access control generally in Splinter, the trust boundary is the OS: who can open the bus is who can participate in the election. Deployments with strict isolation requirements should enforce this at the namespace or cgroup level.

Integration With Governance Standards

Splinter's architecture has direct bearing on compliance with AI accountability and risk management frameworks, including AARM and anticipated parental control and content safety legislation. This section gives an honest account of where the architecture helps, where it creates obligations, and where it delegates responsibility deliberately.

What Splinter Makes Easier

The central compliance value of Splinter is the elimination of the observation gap. In socket-based architectures, governance is necessarily post-hoc: the model produces output, serializes it, transmits it, and only then does a downstream process inspect it. There is an irreducible window during which noncompliant output exists but has not been acted on. Splinter closes this window by construction — the governor reads from the same physical memory the inference engine writes into, and can halt generation before the output is ever considered complete.

This maps directly onto AARM requirements for real-time interruption capability. The shunt-trip mechanism described below is not a policy layer bolted onto an existing architecture — it is a mechanical property of sharing an address space. It can be demonstrated and tested independently of any specific policy configuration.

The bloom label system provides a natural scaffold for audit logging. Each slot carries a 64-bit label mask updated atomically at every state transition. The `evProcessor` reference implementation demonstrates a practical taxonomy: fifteen semantic dimensions with graduated severity levels, fully observable by any process on the bus without data copying. The epoch ordering guarantee means that audit records derived from Splinter are inherently sequenced — there is no ambiguity about whether a governance event preceded or followed a generation event.

The shard election protocol extends this property to memory management: intent is declared, recorded, and observable before it is acted on. This makes the

system's behavior auditable at the infrastructure level, not just the application level.

Observer Pattern As Physics: The Shunt-Trip Method

In electrical engineering, a shunt trip circuit interrupter is a standard GFCI or Arc-Fault interrupter that can also be triggered by an external potential: an environmental sensor, a fire alarm relay, or any signal that would necessitate cutting power independently of the protected circuit's own fault detection.

Splinter's architecture provides this shunt-trip mechanism through direct visibility into the inference process. When several successive token selections match strongly — by cosine similarity and Euclidean distance — against pre-defined third-rail embeddings, signal groups can be immediately pulsed to trigger further governance steps after halting generation. The interrupt is not asynchronous with respect to the output stream. It fires on the same memory the stream is being written into.

Classifying Misalignment In-Flight Instead Of In-Place

A key in Splinter is a Semantic Slot. At T_0, it holds a prompt. At T_1, an inference daemon (*Splainference*) detects the prompt and begins streaming a response into the same slot. At T_2, a Governor is simultaneously reading that same memory stream, calculating the β-NLL loss on the fly.

To use a train analogy: we have a micrometer that understands both magnitude and velocity, and can predict a derailment at almost the instant that circumstances make one possible — not after the train has left the track.

This is a matter of geometric verification, not guesswork; we *know* if "anti" attractor proximity exceeds a defined limit, we can audit that it happened, and we can replay it.

What Splinter Does Not Do, And Why

Splinter provides no authentication and no in-band access control. Any process that can open the bus by name has full read and write access to every slot. This is deliberate — adding an authentication layer would require kernel involvement on every operation, which defeats the architecture's central purpose.

In practice, access control is delegated to the operating system: file permissions on the shared memory object, process isolation via Linux namespaces and cgroups, and deployment topology. Compliance deployments should document these controls explicitly. Auditors evaluating a Splinter-based system should expect to find access controls at the OS layer rather than in the library.

The fixed-geometry constraint has a compliance implication worth stating directly: Splinter can guarantee complete governance coverage only for inference activity that transits the manifold. A client that writes output directly to a socket or

file without going through the bus is outside the observation boundary. This is an integration requirement, not a library limitation, but it must be addressed in any deployment claiming comprehensive coverage.

Finally, `splinter_get_raw_ptr()` provides direct pointer access to slot data, bypassing the library's length and epoch checking. This is documented as sharp-edge behavior and is appropriate for performance-critical read paths that implement their own epoch discipline. Compliance-sensitive code paths should prefer `splinter_get()` or `splinter_get_slot_snapshot()` unless the raw pointer path has been explicitly audited.

Conclusion

Splinter addresses the last-millimeter coordination problem in local LLM deployments by replacing the active-server model with a passive shared-memory manifold. The architecture's three central properties — lock-free atomics aligned to the L1 cache line, seqlock epoch discipline for MRMW consistency, and zero-copy direct pointer access — are not independent optimizations. They are a unified design that makes governance a mechanical consequence of the address space rather than an application-layer concern.

The `evProcessor` reference implementation demonstrates that this is not a theoretical property. A working semantic governance bus with fifteen emotional and epistemic dimensions, graduated severity levels, and full bloom-label routing runs today on modest consumer hardware. The shunt-trip interrupt mechanism fires on the same memory the inference engine writes into, with no socket round-trip and no serialization gap between production and observation.

The Logic Shard protocol extends the same cooperative design philosophy to memory management: rather than fighting the OS scheduler, shards declare their memory intent upfront, elect a sovereign via a deterministic read-only scan, and yield voluntarily when their window expires. The ceiling of 32 shards is not an implementation convenience — it is the same ceiling imposed by the epoch spinlock geometry, and the two constraints compose cleanly.

For AI governance frameworks including AARM, Splinter offers something that post-hoc logging architectures cannot: a verifiable, mechanically-enforced observation boundary. The governor does not receive a report of what the model said. It reads the same memory the model is writing into, in the same moment, with ordering guarantees derived from hardware atomics rather than application protocol.

The Socket And Lock tax is not a fixed cost of doing business with local inference. It is an architectural choice, and Splinter demonstrates that choosing differently is practical on hardware that fits in a laptop bag.

Standards Cited:

- POSIX.1-2017. *posix_madvise — memory advisory information and alignment control.* The Open Group Base Specifications Issue 7.
- llama.cpp. (2024). *LLaMA inference in pure C/C++.* https://github.com/ggerganov/llama.cpp
- Linux `mbind(2)` manual page. *set memory policy for a memory range.* Linux Programmer's Manual.
- AARM Definitions: Autonomous Action Runtime Management Verification https://aarm.dev

Appendix:

- Emotional Valence Processor (Octopus - 3 `evProcessor`) Data Collector Tension Analyzer Currently In Verification Against GDELT Goldstein Index Utilizes semantic base pillar attraction classification through pre-embedded cosine similarity sticking and Euclidean distance magnitude for tripping. https://github.com/splinterhq/prior-art-disclosures

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